Sums and Intersections of Subspaces

Topic.

Vector spaces, subspaces, sums and intersections.

Statement.

Let U_1, U_2, U_3 be subspaces of a vector space. Define

$$V = (U_1 + U_2) \cap (U_1 + U_3) \cap (U_2 + U_3),$$

$$W = ((U_1 + U_2) \cap U_3) + ((U_1 + U_3) \cap U_2).$$

Prove that

$$W \subset V, \qquad V \subset W.$$

Solution.

1. Prove that $W \subset V$.

Let

$$x \in W.$$

By definition of W,

$$\exists a \in (U_1 + U_2) \cap U_3, \qquad \exists b \in (U_1 + U_3) \cap U_2$$

such that

$$x = a + b.$$

Hence

$$a \in U_3, \qquad a \in U_1 + U_2,$$

$$b \in U_2, \qquad b \in U_1 + U_3.$$

First,

$$a \in U_3, \qquad b \in U_2$$

implies

$$x = a + b \in U_2 + U_3.$$

Second,

$$a \in U_1 + U_2, \qquad b \in U_2.$$

Since U_2 is a subspace,

$$b \in U_2 \subset U_1 + U_2.$$

Therefore,

$$x = a + b \in U_1 + U_2.$$

Third,

$$b \in U_1 + U_3.$$

Thus

$$\exists \gamma \in U_1, \quad \exists \delta \in U_3 : \qquad b = \gamma + \delta.$$

Then

$$x = a + b = a + \gamma + \delta = \gamma + (a + \delta).$$

Since

$$a \in U_3, \qquad \delta \in U_3,$$

and U_3 is a subspace,

$$a + \delta \in U_3.$$

Hence

$$x = \gamma + (a + \delta) \in U_1 + U_3.$$

We have shown

$$x \in U_1 + U_2, \qquad x \in U_1 + U_3, \qquad x \in U_2 + U_3.$$

Therefore,

$$x \in (U_1 + U_2) \cap (U_1 + U_3) \cap (U_2 + U_3) = V.$$

Hence

$$W \subset V.$$

2. Prove that $V \subset W$.

Let

$$x \in V.$$

Then

$$x \in U_1 + U_2, \qquad x \in U_1 + U_3, \qquad x \in U_2 + U_3.$$

Therefore, there exist decompositions

$$x = a_1 + a_2, \qquad a_1 \in U_1, \quad a_2 \in U_2,$$

$$x = b_1 + b_3, \qquad b_1 \in U_1, \quad b_3 \in U_3,$$

$$x = c_2 + c_3, \qquad c_2 \in U_2, \quad c_3 \in U_3.$$

We want to represent

$$x = p + q,$$

where

$$p \in (U_1 + U_2) \cap U_3, \qquad q \in (U_1 + U_3) \cap U_2.$$

Put

$$p = c_3, \qquad q = c_2.$$

Then

$$x = c_2 + c_3 = p + q.$$

Now prove that

$$p \in (U_1 + U_2) \cap U_3.$$

We already have

$$c_3 \in U_3.$$

Also,

$$c_3 = x - c_2.$$

Since

$$x = a_1 + a_2,$$

we get

$$c_3 = a_1 + a_2 - c_2 = a_1 + (a_2 - c_2).$$

Here

$$a_1 \in U_1, \qquad a_2, c_2 \in U_2.$$

Since U_2 is a subspace,

$$a_2 - c_2 \in U_2.$$

Therefore,

$$c_3 \in U_1 + U_2.$$

Together with

$$c_3 \in U_3$$

we get

$$p = c_3 \in (U_1 + U_2) \cap U_3.$$

Now prove that

$$q \in (U_1 + U_3) \cap U_2.$$

We already have

$$c_2 \in U_2.$$

Also,

$$c_2 = x - c_3.$$

Since

$$x = b_1 + b_3,$$

we get

$$c_2 = b_1 + b_3 - c_3 = b_1 + (b_3 - c_3).$$

Here

$$b_1 \in U_1, \qquad b_3, c_3 \in U_3.$$

Since U_3 is a subspace,

$$b_3 - c_3 \in U_3.$$

Therefore,

$$c_2 \in U_1 + U_3.$$

Together with

$$c_2 \in U_2$$

we get

$$q = c_2 \in (U_1 + U_3) \cap U_2.$$

Thus

$$x = p + q$$

where

$$p \in (U_1 + U_2) \cap U_3, \qquad q \in (U_1 + U_3) \cap U_2.$$

Hence

$$x \in W.$$

Therefore,

$$V \subset W.$$

Since

$$W \subset V \qquad \text{and} \qquad V \subset W,$$

we conclude that

$$V = W.$$